

ST CLOUD, MN, USA

WMO: 726550

Lat: 45.543N

Lon: 94.051W

Elev: 1018

StdP: 14.16

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14926

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.8	-11.4	-23.0	1.6	-15.6	-18.4	2.1	-10.0	26.4	11.9	24.1	13.2	5.3	310	0.598

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.7	89.3	72.6	86.3	70.7	83.5	68.9	76.0	85.1	73.9	82.8	71.9	80.1	12.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.1	127.7	81.9	70.9	118.2	79.1	69.0	110.6	77.1	40.2	84.6	38.1	82.9	36.2	79.8	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	20.4	18.4	-24.4	94.7	6.6	2.7	-29.2	96.6	-33.0	98.2	-36.7	99.7	-41.6	101.7
			-23.4	79.6	4.5	2.2	-26.6	81.1	-29.3	82.4	-31.8	83.6	-35.1	85.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.5	12.0	16.2	29.5	43.8	56.2	66.3	70.9	68.2	60.5	46.7	31.8	18.3
	DBStd	22.80	13.56	12.79	12.59	9.98	8.45	6.72	5.73	5.55	8.45	9.07	10.64	12.16
	HDD50	4758	1178	946	643	234	34	1	0	0	13	174	551	984
	HDD65	8350	1643	1366	1102	638	295	61	13	30	185	571	997	1449
	CDD50	2383	0	0	6	47	227	491	646	563	327	71	5	0
	CDD65	503	0	0	0	1	23	102	195	128	50	4	0	0
	CDH74	5029	0	0	1	29	335	1032	1934	1176	478	44	0	0
	CDH80	1522	0	0	0	6	103	305	655	322	125	6	0	0
	WSAvg	8.3	8.6	8.7	8.9	9.6	9.0	7.7	6.8	6.4	7.6	8.7	8.7	8.4
	PrecAvg	29.1	0.7	0.8	1.6	2.7	3.7	4.3	3.8	3.7	2.8	2.6	1.4	0.9
Precipitation	PrecMax	37.0	2.1	1.5	4.3	8.0	8.7	8.7	7.6	8.0	6.0	6.3	4.0	2.3
	PrecMin	20.8	0.1	0.1	0.4	0.7	1.2	2.0	1.4	0.5	0.3	0.7	0.0	0.1
	PrecStd	4.1	0.5	0.4	0.9	1.6	1.9	1.4	1.7	1.9	1.6	1.4	1.1	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.7	49.7	66.8	77.8	89.4	92.2	94.2	91.2	88.2	79.9	65.0	47.0
		MCWB	36.8	44.0	55.5	59.2	67.2	73.7	76.1	74.6	70.5	63.9	54.2	41.5
	2%	DB	36.9	42.1	58.5	71.2	82.4	87.0	89.8	86.8	83.7	72.5	57.6	40.4
		MCWB	33.7	37.5	50.4	54.0	63.3	70.1	74.4	72.0	69.3	59.8	49.3	36.7
	5%	DB	33.7	37.6	52.0	66.2	77.2	83.3	86.5	83.4	79.8	66.5	51.8	36.8
		MCWB	31.8	34.8	45.7	51.3	60.5	67.6	72.0	69.6	66.6	55.7	44.9	34.2
	10%	DB	30.8	34.3	46.1	61.3	72.4	80.1	83.5	80.5	75.4	62.0	47.2	34.0
		MCWB	28.9	32.0	40.9	49.0	58.1	65.9	70.0	68.3	64.7	53.7	42.2	32.0
	0.4%	WB	37.2	44.4	59.3	61.5	71.4	76.6	80.0	77.9	74.8	66.5	56.7	42.4
		MCDB	41.7	49.7	64.1	74.5	84.5	87.8	88.8	86.2	82.8	74.5	62.1	45.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.2	38.3	51.6	56.7	66.4	73.3	77.0	74.9	71.5	62.0	50.8	37.1
		MCDB	36.6	41.5	57.3	67.2	76.0	83.5	86.4	83.3	79.9	69.3	56.7	40.0
	5%	WB	31.9	34.9	46.0	53.1	63.4	70.8	74.6	72.7	69.3	58.2	45.9	34.5
		MCDB	33.9	37.6	52.0	63.9	74.1	80.1	83.5	80.4	76.2	65.3	51.1	36.6
	10%	WB	29.0	32.3	41.2	50.0	60.3	68.3	72.5	70.4	66.6	54.2	42.2	32.2
		MCDB	30.8	34.5	46.3	60.1	69.8	76.6	80.9	77.3	73.5	60.7	46.7	34.1
Mean Daily Temperature Range	5% DB	MDBR	16.9	18.2	17.9	20.9	22.3	21.1	21.7	21.6	21.9	19.2	16.2	15.1
		MCDBR	17.7	19.1	24.4	29.9	30.6	25.6	24.5	24.7	26.4	27.3	24.1	18.0
		MCWBR	15.1	14.9	16.2	16.5	14.2	11.6	11.4	11.7	13.3	15.6	16.1	14.5
	5% WB	MCDBR	17.4	17.4	22.9	25.7	24.9	21.7	21.8	20.8	22.0	23.0	22.2	16.1
		MCWBR	14.9	14.2	16.0	15.7	13.6	11.5	11.5	11.8	12.7	14.8	15.7	13.5
Clear-Sky Solar Irradiance	taub		0.263	0.276	0.312	0.369	0.402	0.391	0.385	0.399	0.367	0.314	0.285	0.260
	taud		2.401	2.359	2.379	2.299	2.261	2.337	2.376	2.336	2.398	2.537	2.509	2.433
	Ebn at Noon		272	290	292	282	275	278	278	270	270	273	263	262
	Edh at Noon		23	29	33	39	42	39	37	37	32	24	20	20
All-Sky Solar Radiation	RadAvg		543	889	1218	1485	1704	1889	2012	1676	1288	809	514	377
	RadStd		66	93	108	131	134	131	69	132	93	113	48	49

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (65 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page